

Codex / Lean formalization handoff

An exact candidate below $7/8$

$$\sigma_0 = \frac{437479}{500000} = 0.874958$$

Prepared for Steven Creech | 8 October 2026

Task. Formalize the parameter improvement in Lean 4 with mathlib. First prove the arithmetic certificate unconditionally. Then formalize the analytic implication with every imported assumption visible. The ultimate target concerns the actual Riemann zeta function at every height:

$$\forall s \in \mathbb{C}, \quad \Re s > \frac{437479}{500000}, \quad s \neq 1 \implies \zeta(s) \neq 0.$$

The improvement in the boundary is exactly $21/500000 = 0.000042$.

Current proof status. This PDF contains an exact rational parameter certificate and a written *conditional* analytic derivation. It is not a completed Lean proof, and it does not independently verify the analytic lemmas of the source manuscript. No Lean compiler was available in the preparation environment. All Lean listings below are uncompiled starting points; mathematical arithmetic checks were performed with exact fractions.

Deliverable	What a successful formalization establishes
Arithmetic certificate	The displayed geometry, polynomial identities, and strict inequalities, without analytic assumptions.
Conditional assembly	The improved region follows from explicitly stated, fully quantified source estimates and their applicability to the new probe.
Unconditional theorem	Every analytic input and its connection to the actual zeta function has also been proved or imported as a proved Lean theorem.

Instructions to the formalizing agent. Preserve the strict boundary and exact rational constants. Do not replace an analytic obligation by an axiom, an assumed improved zero bound, or an opaque “bridge” predicate and call the result unconditional. If an obligation cannot be proved, record its precise statement and dependency and continue with the independent arithmetic modules. Inspect both theorem types and their axioms.

How to use this document. Sections 1–8 preserve the mathematical audit. Appendices A–H specify the implementation order, exact definitions, missing row formulas, source dependencies, quantifier order, zeta interface, and acceptance criteria. The quartic squared-theta investigation is not an input to this candidate and must not be substituted for the source moments.

Source required for the analytic stage. The 199-page, 30 September 2026 manuscript linked in Section 7 is required in addition to this handoff. Its exact local PDF fingerprint and the relevant source locations are in Appendix A. If that source cannot be obtained, complete the arithmetic stage and report the analytic stage as blocked by the missing source statements.

A parameter audit below seven eighths

A conditional derivation of the threshold 0.874958

Research working note prepared for Steven Creech

8 October 2026

Abstract

Taking the analytic identities and estimates in the supplied OpenAI manuscript as inputs, we modify its compensated probe and derive the threshold $437479/500000 = 0.874958$. The low estimate is recomputed, the principal Euler correction is extended, and the intermediate-row inequality is certified by an explicit positive polynomial. This note is a conditional derivation and audit, not an independent verification of the source manuscript or a claim of an independently established new zero-free theorem.

1 Status and precise inputs

Let $[M]$ denote the 30 September 2026 manuscript, *The Quasi-Riemann Hypothesis: A Zero-Free Half-Plane* $\Re s > 7/8$. Numbered lemmas and equations below refer to its 199-page version. We use its original family of finite-order Hecke characters over $\mathbb{Q}(\sqrt{-3})$, its supremum β^* of zero real parts, and its prime-ideal norm convention.

The inputs are the manuscript's initial conclusion $\beta^* \leq 7/8$; its exact reflection, completion, and compensated-probe identities; the fixed-part reflected estimate of Lemma 14.3; Proposition 15.2; the witness and moment estimates of Sections 8, 17, and 18, especially Lemmas 17.1–17.2, 17.6, and 18.1; and their stated uniformity and smooth-weight bounds. The present calculation does not independently prove those inputs. In particular, a certificate for an exponent inequality is not a certificate for all the analytic identities used to obtain it.

Conditional proposition 1. *Assuming those inputs and their stated uniformity, the modified argument below gives $\beta^* \leq 437479/500000$. Proposition 11.3 of $[M]$ then transfers the same strict zero-free half-plane to the Dirichlet family, including $\zeta(s)$.*

Choose the following rational parameters:

$$\ell = \frac{62563}{375000} = \frac{1}{6} + \frac{21}{125000}, \quad b = \frac{1}{8},$$
$$l_x = \frac{132781}{375000}, \quad l_y = \frac{22457}{46875}, \quad h = \frac{50797}{62500}, \quad \sigma_0 = \frac{437479}{500000}.$$

They satisfy $l_x + l_y + \ell = 1$, $l_y - l_x = b$, and $h = 1 - l_x + \ell$. The signal exponent is unchanged:

$$C(s) = s + \frac{l_x}{2} - 1 + \frac{h}{6} = s - \frac{11}{16}.$$

The low exponent is $C(\sigma_0) = 93729/500000$. The improvement in the boundary over $7/8$ is exactly $21/500000$.

2 Recomputing the low bound

Use the finite physical expression (12.5), with the new positive slot lengths summing to ℓ . For a rescaled subset with total length d , set

$$M = 1 - \ell, \quad M' = M - 2d, \quad \ell' = \ell - d, \quad 0 \leq d \leq \ell.$$

The original statement of Lemma 15.1 cannot simply be reused: its numerical loss must be recomputed. Its proof, using Lemma 14.3, gives the two branch exponents

$$M' - \Delta_H + O(\varepsilon), \quad M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H}{4} + O(\varepsilon).$$

Here $\Delta_H \geq -O(1/\log Z)$, and

$$1 + 3\ell' - 2M' = 5\ell - 1 + d.$$

Thus, with the original row-ball scale $Q \asymp Z^{M'}$, the completed-row squared norm is bounded by

$$\sum_{0 < N(m) \ll Q} |B_m^J|^2 \ll Z^{M' + (5\ell - 1 + d)_+/4 + \varepsilon}. \quad (1)$$

This retains the same masks, joint profiles, and fixed-part summation as the printed proof.

For the present rational parameters, the three scale conditions used in Proposition 15.3 remain strict: $l_x - \ell > 0$, $l_y - \ell > 0$, and

$$l_y - \ell - \frac{11b}{6} > 0.$$

Hence Proposition 15.2 has the same dominating Gram term as in that proof. Cauchy–Schwarz contributes half the extra exponent in (1). Summing the rescaled prime tuples and their coefficients supplies the original factor Z^{-d} relative to the unrescaled term. The complete extra exponent is therefore

$$f(d) = -d + \frac{(5\ell - 1 + d)_+}{8}.$$

Its slopes are -1 and $-7/8$. Since $\ell < 1/5$, $f(0) = 0$, and consequently $f(d) \leq 0$ for every rescaled subset. This is the point that prevents the extra row-norm loss from spoiling the perturbation.

After division by the same logarithmic prime normalizer as in Section 20,

$$|J_\eta(Z)| \ll_{\eta, \varepsilon} Z^{l_x/2 + b/12 + \varepsilon} = Z^{C(\sigma_0) + \varepsilon}. \quad (2)$$

The fixed character is denoted η here, as in the manuscript; it is not the parameter increment $21/125000$.

3 The moment parameter and row counts

Assume for contradiction that $\sigma_0 < \beta^* \leq 7/8$. Keep $\kappa = 3/4$ in Lemma 18.1. Its hypothesis $\beta^* \leq (1 + \kappa)/2$ is satisfied. Thus no extension of that moment theorem below its printed parameter range is needed.

For a non-floor bin let $\delta = 2a - 1$, $q = x\delta$, so

$$\frac{1}{50} < \delta \leq \frac{3}{4}, \quad 0 \leq x \leq \frac{1}{2}.$$

Re-run the proof of Proposition 19.2 with this fixed κ . Its plain capacity is exactly $2(1 - 2m)/9$, so there is no $\Delta/4$ capacity-comparison loss. One does not invoke the proposition literally with its original assumption $\beta^* > 7/8$.

The two ideal row counts, for witness parameter $1 \leq t \leq 3/2$, are

$$R_{\text{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x} \left(\frac{3}{2} - t \right), \quad L(t) = 1 - \delta + \left(\frac{5}{6} - \delta \right) (t - 1),$$

where

$$D_x = 3 - \frac{17x}{9}, \quad P_x = \left(2 - \frac{8x}{9} \right) (1 - x).$$

Balance them by setting

$$J = \left(\frac{5}{6} - \delta \right) D_x + \delta P_x, \quad t = 1 + \frac{\delta P_x}{2J}, \quad R = 1 - \delta + \frac{(5/6 - \delta)\delta P_x}{2J}.$$

Here $J \geq 35/54$ and $J \leq 5/2$. The maximum required inverse capacity is $7/37$, and

$$\frac{\ell}{h} - \frac{7}{37} = \frac{181357}{11276934} > 0.$$

Consequently the finite slot mesh can be chosen with a fixed positive supply margin. The original small-capacity treatment and smooth witness bounds continue to give a count $U^{R+\varepsilon}$ with the same type of fixed polynomial height factor.

4 An exact certificate for the high endpoint

Relative to $C(\sigma_0)$, the ideal intermediate-row endpoint exponent is

$$E = a - \sigma_0 + h \left(R + \frac{\delta}{2} - \frac{1}{6} \right) - a l_y - \frac{\ell}{2} + x \delta \ell, \quad a = \frac{1 + \delta}{2}. \quad (3)$$

Put $y = 1/2 - x$. Direct rational expansion gives

$$81,000,000(-2JE) = A(y)\delta^2 + B(y)\delta + C_0(y),$$

where

$$\begin{aligned} A(y) &= 38265876 + 98256048y + 70521168y^2 + 24024192y^3, \\ B(y) &= -29594760 - 47116180y - 3252520y^2, \\ C_0(y) &= 5722975 + 5258950y. \end{aligned}$$

The identity is checkable by clearing the denominator $2J$ in (3). Moreover,

$$\begin{aligned} 4AC_0 - B^2 &= 128787346800 + 265436861138400y \\ &\quad + 1268808584874800y^2 + 1726935951112000y^3 \\ &\quad + 494789211723200y^4. \end{aligned}$$

All coefficients are positive. Therefore $A > 0$ and

$$A\delta^2 + B\delta + C_0 = A \left(\delta + \frac{B}{2A} \right)^2 + \frac{4AC_0 - B^2}{4A} > 0.$$

This proves $E < 0$ throughout the required rectangle, without a floating-point optimization or a sampled-grid inference.

For an explicit uniform margin, on $0 \leq y \leq 1/2$ one has $A(y) \leq 108027216$ and $2J \leq 5$. Thus

$$E \leq -\gamma, \quad \gamma = \frac{35774263}{48612247200000} > 7.35 \cdot 10^{-7}. \quad (4)$$

This deliberately conservative bound suffices for all subsequent small losses.

The other row regimes retain much larger margins. Substitution in their displayed formulas gives

Regime	Ideal relative exponent upper bound
Floor bin, at $d = h$	$-105343/18750000$
$d \leq 1/2$, unselected count	$-1872449/75000000$
Tiny rows, $d_{\min} = 1/100$	$-122913/1562500$

In the unselected count use $R = 76/75 - 2\delta/3$. The endpoint slopes are positive and less than two. Choose an extension $0 < \zeta < \gamma/8$ sufficiently small to retain prime supply. The interval $h < d \leq h + \zeta$ then costs less than $\gamma/4$. For larger rows the original absolute-line argument with sufficiently large fixed z_∞ applies. The inequality $5\ell - h = 8033/375000 > 0$ provides ample supply margin for this extension.

5 Euler correction and contour audit

The printed lower bound $7/8$ in the principal Euler region is not a singularity threshold. Write

$$\mathcal{D}_2(\sigma) = \{\Re s \geq \sigma, \Re z \geq 33/200, \Re w \geq 19/20\}.$$

The local formulas in Lemma 7.1 give, for $\sigma \geq 3/4$,

$$H_p - 1 = O(Q^{-c(\sigma)}) \ (p \nmid u), \quad H_p - 1 = O(Q^{-7/10}) \ (p \mid u),$$

where

$$c(\sigma) = \min \left(6\sigma - \frac{301}{100}, \sigma + \frac{47}{50}, \frac{97}{50} \right) > 1.$$

At $\sigma = 3/4$ this is $149/100$. The local geometric denominators stay separated from zero. Normal convergence and the divisor-product bound give the same holomorphic correction and $N(u)^\varepsilon$ bound. For $u = 1$, increasing the fixed excluded-prime cutoff gives

$$H_\eta(s) = 1 + O(P_0^{-49/100}), \quad \sup_{\Re s > \sigma_0} |H_\eta(s) - 1| \leq \frac{1}{2}.$$

The principal slot multiplier in (16.8) satisfies

$$B_p = -1 + O(Q^{-b_0(\sigma)}), \quad b_0(\sigma) = \min \left(\sigma, \frac{99}{100}, 5\sigma - \frac{301}{100}, \frac{47}{50} \right).$$

For $\sigma \geq 3/4$, $b_0(\sigma) \geq 37/50$. The same positive prime-window normalizer therefore works, with any fixed

$$0 < \kappa_P < \frac{37}{50} \min_i \ell_i.$$

The quotient-free tuple formula (16.3) supplies the correction's polynomial height bounds on the extended region.

The nonprincipal fixed-bin contours use the unchanged first Euler region. The principal residue contours use the extended second region; their savings remain $l_y/20$, $h/600$, and κ_P . On the small-row contour the first-region parameter should be $1/4$, replacing the printed $3/8$. The selected factors remain $O(1)$ off the row and $O(Q^{1/2})$ on it. Thus this change does not require a new character-sum estimate.

6 Order of choices and conditional closure

Let $\Delta = \beta^* - \sigma_0 > 0$. First fix the rational geometry and a positive high margin smaller than (4). Fix ζ , the small-capacity decrements, amplitude bins and slot mesh, and choose their total exponent cost smaller than a prescribed fraction of that margin. Reduce the low loss so that it is less than $\Delta/4$. After the finite slot system and κ_P are fixed, choose the detector width e and principal loss ε_{pr} sufficiently small that

$$(1 + h)e + \varepsilon_{\text{pr}} < l_y/80, \quad e + \varepsilon_{\text{pr}} < h/2400, \quad e + \varepsilon_{\text{pr}} < \kappa_P/4.$$

These choices may depend on the common supremum and fixed geometry, but not on an individual target character.

For each target, retain the manuscript's common-height procedure: after the finitely many internal seminorm and height orders are fixed, choose its height exponent sufficiently small, then the external tail orders. This is the order used by Lemma 11.1 and Section 20.6. It absorbs the fixed polynomial height factors into the reserved positive margins; no uniform constant over all conductors is required. The prime normalizer costs only an arbitrarily small power of Z .

The resulting estimates have the form

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega}, \quad |J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta^*)-s_1},$$

with $0 < \omega < \Delta$ and $s_1 > 0$ independent of the target. The signal and physical probe remain independent of the later analysis height. Proposition 2.1, already stated for every $\sigma_0 \in (1/2, 1)$, contradicts $\beta^* > \sigma_0$. This closes the modified argument *relative to the source inputs listed on page 1*.

The work completed here verifies the parameter-dependent modifications and provides a checkable high-exponent certificate. It does not constitute an independent audit of the source's long reflection and recursive moment arguments, and should not be circulated as an unqualified proof of a new zero-free theorem.

7 What limits further progress

There is an explicit obstruction to substantially improving the present row envelope by changing only b and ℓ . At $x = 1/2$, put

$$\delta_* = \frac{49 - \sqrt{921}}{48} = 0.388583712\dots$$

Then $R(\delta_*, 1/2) = 2/3$. In the clean low regime, where $\sigma_0 = 11/12 - \ell/4$, the endpoint exponent becomes independent of b :

$$E = \frac{\delta_*}{2} - \frac{5}{12} + \frac{3}{4}(1 + 2\delta_*)\ell.$$

Strict negativity forces

$$\ell < \ell_* := \frac{5 - 6\delta_*}{9(1 + 2\delta_*)} = 0.16683838746999\dots$$

Consequently this fixed- κ envelope cannot reach a boundary at or below

$$\frac{11}{12} - \frac{\ell_*}{4} = 0.87495706979917\dots$$

This is a limitation of these particular estimates and geometry, not a theorem ruling out better zero-free regions. Numerical optimization suggests the obstruction is nearly sharp for this two-parameter tuning, but that numerical attainment is not needed for the exact candidate proved above.

Where a stronger estimate is needed. The restrictive bin is neither near the largest possible zero real part nor at the floor: it has $a = (1 + \delta_*)/2 \approx 0.69429$ and maximal prime amplitude $x = 1/2$. Improving the inverse-witness count, the plain-witness capacity, or their joint treatment in this bin would affect the actual bottleneck. Merely extending the principal Euler region to $3/4$ does not do so.

Additional low-side work, not used in the candidate. A separate audit suggests a sharper aggregate count for frozen powerful row parts: inactive good primes have sixth-power valuations, which permits $Z^{O/6+2A_0/3+\varepsilon}$ in place of $Z^{O/2}$. Combining this with the fixed-part reflected estimate suggests the stronger low threshold

$$\frac{11}{12} - \frac{\ell}{4} + \frac{(4\ell - 1)_+}{6}.$$

Its low-side minimum is $41/48$, but the high constraint above is already binding far earlier. This strengthening is not required for 0.874958 and should receive its own proof review before use elsewhere.

Source and reproducibility. The source manuscript is available from the official repository:

[OpenAI, The Quasi-Riemann Hypothesis \(30 September 2026\).](#)

This revision tightens the earlier candidate 0.87496 to 0.874958 . All decimal values in the candidate proof are accompanied by rational definitions. The high certificate requires only polynomial multiplication and completion of a square. Its coefficient identity and discriminant were checked with exact rational arithmetic; numerical optimization was used only to choose promising parameters and explore the limitation described above.

8 The saving and the zeta-specific question

The arithmetic comparison is

$$\frac{7}{8} = 1 - \frac{1}{8}, \quad \frac{3}{4} = 1 - \frac{1}{4}, \quad \frac{7}{8} - \frac{3}{4} = \frac{1}{8}.$$

Thus reaching $3/4$ asks for an additional one-eighth saving relative to a bound already giving $7/8$. An additional one-quarter would be the wrong comparison. The result above improves the boundary by only $21/500000$; it does not establish $5/6$ or $3/4$.

Why a linear quartic moment does not supply that increment. At the original geometry, the completed-row norm in the manuscript is already linear in its row scale. The separately derived linear quadratic-row moment for quartic Gauss coefficients therefore does not sharpen that exponent. There is also a change of field and coefficients: the manuscript's quadratic-cubic estimate is over $\mathbb{Q}(\sqrt{-3})$ and permits arbitrary bounded coefficients with moving coprimality masks; the quartic calculation is over $\mathbb{Q}(i)$ with particular Gauss coefficients. A direct replacement would require an estimate matching those inputs.

What restricting the target to zeta changes. Setting the target character to 1 does not set the moving internal characters χ_u to 1. The reflected expression (7.13) still contains

$$\frac{L^S(w, \chi_u)}{L^S(s, \overline{\chi_u})}.$$

Applying the numerator functional equation and writing $v = s + w - 1$ produces a ratio proportional to

$$\frac{L^S(s - v, \overline{\chi_u})}{L^S(s, \overline{\chi_u})}.$$

At $v = 0$ this is identically 1 after removable continuation. That observation alone does not control the rest of the contour. For example, if the local Euler extraction supplies a factor $\zeta_F^S(1 + v)$ with residue κ_S , the Laurent expansion is

$$\zeta_F^S(1 + v) \frac{L^S(s - v)}{L^S(s)} = \frac{\kappa_S}{v} - \kappa_S \frac{(L^S)'(s)}{L^S(s)} + c_S + O(v).$$

Although the v -residue has lost the reciprocal L -function, the regular coefficient retains its zero poles. A new estimate is still required before this manipulation improves a zero-free region.

The next substantive target. The exact endpoint certificate isolates the restrictive row bin. A useful further result would improve its inverse and plain witness counts jointly, or construct a new completed probe retaining quartic coefficient correlations through reflection. The location of a theta-series pole at $3/4$ is not itself the nonpolar bound required by either route.

A Formalization scope, source fingerprint, and modules

Read this before creating Lean declarations. The intended theorem is conditional on the analytic source until those inputs are proved. The arithmetic stage is self-contained in this PDF. The analytic stage requires the actual Hecke family over $F = \mathbb{Q}(\sqrt{-3})$, the physical compensated probe, and the complete statements of the source estimates. An arbitrary function named `zeta` is not an implementation of the target.

The source $[M]$ is the 199-page PDF at

[openai/math](#), preprint dated 30 September 2026.

SHA-256 of the exact copy used for this audit:

8fe93046f8cf5ef1ba5969c89addc02d76311adc4ee907509ff9cd96f7ec99e7

This identifies the audited copy; the mutable repository URL is not a commit pin. If a downloaded copy differs, check the numbered statements before reusing references. Printed and PDF page numbers agree in the audited copy. Record the source hash, Lean toolchain, and exact mathlib commit in the project.

Suggested module	Contents and prerequisites
<code>Parameters.lean</code>	Exact constants, geometry, signal, improvement.
<code>LowArithmetic.lean</code>	Positive-part inequality for rescaled subsets.
<code>RowArithmetic.lean</code>	D_x, P_x, J, t, R ; positivity and balancing.
<code>HighCertificate.lean</code>	Polynomial identity, discriminant, uniform $-\gamma$.
<code>OtherRegimes.lean</code>	Floor, unselected, tiny, extension, Euler exponents.
<code>Arithmetic.lean</code>	Bundle the preceding results as one certificate.
<code>LateHeight.lean</code>	Generic exponent/quantifier lemmas; no Hecke inputs.
<code>AnalyticInputs.lean</code>	Explicit source statements about actual objects.
<code>ModifiedProbe.lean</code>	Reprove changed low, row-count, and contour steps.
<code>Continuation.lean</code>	Common signal, Mellin continuation, supremum argument.
<code>ZetaTransfer.lean</code>	Actual family-to-zeta connection; conditional main theorem.

The first six modules can be developed without the automorphic machinery. Do not make their compilation depend on admitted analytic files. The source’s initial 7/8 theorem is an analytic input, not a consequence of their arithmetic. The limitations and exploratory strengthening in Section 7 are not needed.

Notation discipline. In the Lean specification, use u for a rescaled subset length, r for the physical row exponent in $U = Z^r$, and $\delta = 2a - 1$ for the dynamic bin. The witness parameter $t \in [1, 3/2]$ is not an imaginary part. Use `rowJ` for the positive denominator; reserve $J_\eta(Z)$ for the physical probe. Use `geomH` for h , distinct from the holomorphic correction H_η .

B Exact Lean definitions and the first arithmetic goals

The following is an **uncompiled starter specification**. Use explicit `Real` types; untyped natural-number division would change the problem. Initially `import Mathlib` is convenient. Narrow imports after a successful build. All later formulas use these definitions.

```
import Mathlib
namespace QuasiRiemann
noncomputable section

def sigma0 : Real := (437479 : Real) / 500000
def ell : Real := (62563 : Real) / 375000
def b : Real := (1 : Real) / 8
def lx : Real := (132781 : Real) / 375000
def ly : Real := (22457 : Real) / 46875
def geomH : Real := (50797 : Real) / 62500
def gammaSave : Real := (35774263 : Real) / 48612247200000
def signal (s : Real) := s + lx / 2 - 1 + geomH / 6
def lowExtra (u : Real) := -u + max 0 (5 * ell - 1 + u) / 8

def rowD (x : Real) := 3 - 17 * x / 9
def rowP (x : Real) := (2 - 8 * x / 9) * (1 - x)
def rowJ (d x : Real) := (5 / 6 - d) * rowD x + d * rowP x
def witnessT (d x : Real) := 1 + d * rowP x / (2 * rowJ d x)
def rowR (d x : Real) :=
  1 - d + (5 / 6 - d) * d * rowP x / (2 * rowJ d x)
def binA (d : Real) := (1 + d) / 2
def highE (d x : Real) :=
  binA d - sigma0 + geomH * (rowR d x + d / 2 - 1 / 6)
  - binA d * ly - ell / 2 + x * d * ell

theorem exact_improvement :
  (7 : Real) / 8 - sigma0 = 21 / 500000 := by
  norm_num [sigma0]

end
end QuasiRiemann
```

Prove named lemmas for every identity in Section 1, including $\sigma_0 = 11/12 - \ell/4$, $C(s) = s - 11/16$, and $l_x/2 + b/12 = C(\sigma_0)$. The exact positive geometry margins are

$$l_x - \ell = \frac{11703}{62500}, \quad l_y - \ell = \frac{39031}{125000}, \quad l_y - \ell - \frac{11b}{6} = \frac{62311}{750000}.$$

Also prove $1/2 < \sigma_0 < 7/8$, $0 < \ell < 1/5$, $h > 0$, $\gamma > 0$, and the two supply identities in Sections 3–4.

Low inequality. For every real $u \geq 0$,

$$\max(0, 5\ell - 1 + u) \leq u, \quad \text{lowExtra } u \leq -\frac{7}{8}u \leq 0.$$

Use $5\ell - 1 \leq 0$, `max_1e`, and linear arithmetic. The original application also has $u \leq \ell$, but the displayed arithmetic is stronger. This does not prove the masked row norm or the factor Z^{-u} .

C The high certificate as explicit Lean lemmas

Work on the closed real rectangle $0 \leq \delta \leq 3/4$, $0 \leq x \leq 1/2$. The analytic non-floor bins have $\delta > 1/50$; proving the closed rectangle is convenient and stronger. First prove

$$\frac{37}{18} \leq D_x \leq 3, \quad \frac{7}{9} \leq P_x \leq 2, \quad D_x - P_x = 1 + x - \frac{8}{9}x^2 \geq 0, \quad \frac{35}{54} \leq J \leq \frac{5}{2}.$$

For example, $J = (5/6)D_x - \delta(D_x - P_x)$ gives the upper bound, and $J = (5/6)P_x + (5/6 - \delta)(D_x - P_x)$ gives the lower bound. Record $J \neq 0$ and $D_x \neq 0$ before clearing denominators. Prove $1 \leq t \leq 3/2$, the equality of the two row counts, and

$$1 - \delta \leq R \leq 1 - \delta + \frac{5/6 - \delta}{2}.$$

Define `polyA`, `polyB`, `polyC` by the coefficients in Section 4, and set

$$P(\delta, y) = A(y)\delta^2 + B(y)\delta + C_0(y), \quad D(y) = 4A(y)C_0(y) - B(y)^2.$$

Use a name such as `discPoly` for $D(y)$ to avoid confusion with D_x . The essential lemma statements are

$$\begin{aligned} 81000000(-2J(\delta, x)E(\delta, x)) &= P(\delta, 1/2 - x) & (J \neq 0), \\ D(y) &= 128787346800 + 265436861138400y \\ &\quad + 1268808584874800y^2 + 1726935951112000y^3 + 494789211723200y^4, \\ 4A(y)P(\delta, y) &= (2A(y)\delta + B(y))^2 + D(y). \end{aligned}$$

The first is a rational-function identity: use `field_simp` with the nonzero denominator proof, then `ring`. The last two are polynomial identities proved by `ring`. Do not use numerical sampling.

For $0 \leq y \leq 1/2$ prove

$$0 < A(y) \leq 108027216, \quad D(y) \geq 128787346800.$$

For the upper bound on A , use $y^2 \leq 1/4$ and $y^3 \leq 1/8$. The square identity first gives $P > 0$, then

$$128787346800 \leq 4A(y)P(\delta, y) \leq 4(108027216)P(\delta, y).$$

Combine this with the first identity and $0 < 2J \leq 5$ to obtain

$$E(\delta, x) \leq -\frac{128787346800}{4 \cdot 108027216 \cdot 81000000 \cdot 5} = -\gamma.$$

This avoids completing a square with a new variable denominator $A(y)$. For tractable `nlinarith` calls, keep A, P, J, E as local abbreviations after proving the identities.

Export `high_margin` with the four rectangle hypotheses and conclusion `highE d x <= -gammaSave`. Also prove $735/10^9 < \gamma$ using exact rational arithmetic. A proof of $E < 0$ at finitely many points is not a substitute for this universally quantified lemma.

D Other regimes, extension, and Euler arithmetic

The original table in Section 4 is not enough for formalization without its defining expressions. Set $a = (1 + \delta)/2$ and define

$$E_{\text{gen}}(r, \delta, q, R) = a - \sigma_0 + h(17/50 - 1/6) - al_y - \ell/2 + q\ell \\ + r(R + \delta/2 - 17/50).$$

This exponent is measured relative to $C(\sigma_0)$. Prove $E(\delta, x) = E_{\text{gen}}(h, \delta, x\delta, R(\delta, x))$. The amplitude term is $q\ell$, *not* rql .

Floor. At $\delta = 1/50$, $R = 1$, $q \leq 1/100$, $r \leq h$,

$$E_{\text{gen}}(r, 1/50, q, 1) \leq E_{\text{gen}}(h, 1/50, 1/100, 1) = -\frac{105343}{18750000}.$$

Use $\ell > 0$ and row slope $67/100 > 0$.

Unselected rows. For $0 \leq \delta \leq 3/4$, $q \leq \delta/2$, $r \leq 1/2$, and $R = 76/75 - 2\delta/3$, the slope is $101/150 - \delta/6 > 0$. The upper endpoint equals

$$E_{\text{gen}}(1/2, \delta, \delta/2, R) = -\frac{8263987}{37500000} + \frac{195407}{750000}\delta \leq -\frac{1872449}{75000000}.$$

Tiny rows. Their separate expression is

$$h(17/50 - 1/6) - l_y/2 + 2/100 = -\frac{122913}{1562500}.$$

This saving is relative to $C(\beta^*)$, as in $[M]$, Lemma 10.6 and Section 20.3. Do not silently relabel it as relative to $C(\sigma_0)$.

Extension beyond h . For the selected R , the bounds in Appendix C give $0 < R + \delta/2 - 17/50 < 2$. One explicit admissible choice is $\zeta = \gamma/16$. Check by rational arithmetic that

$$0 < \zeta < \gamma/8, \quad \zeta < 5\ell - h, \quad \frac{\ell}{h + \zeta} > \frac{1}{5} > \frac{7}{37}.$$

Thus extending to $h + \zeta$ costs at most $2\zeta = \gamma/8$. The large-row absolute-line argument is an analytic obligation, not covered by this finite certificate. Its exponent is $B_0 + (h + \zeta)(1 + \varepsilon) - \zeta z_\infty + \varepsilon$, where $B_0 = l_x/2 + 1 + l_y$; choose fixed z_∞ large enough.

Euler arithmetic. Define $c(s)$ and $b_0(s)$ as the nested minima in Section 5. Prove for every real $s \geq 3/4$ that

$$c(s) \geq 149/100 > 1, \quad b_0(s) \geq 37/50.$$

Use `le_min` and linear arithmetic. These exponent inequalities do not themselves prove normal convergence, denominator separation, contraction, or polynomial growth of the actual Euler correction.

E Analytic input ledger and mandatory parameter changes

All source locations below refer to [M]. An imported assumption must contain its full mathematical statement about the actual objects, not merely a name. Use explicit arguments or a fully displayed record; avoid invisible typeclass instances supplying unproved analytic assertions.

Source statement	Role and restrictions that must survive
Theorem 1.1; p.197	Initial $\beta^* \leq 7/8$ for the full Hecke family.
Prop.2.1; pp.8–9	Common-signal continuation; variable boundary in $(1/2, 1)$.
Prop.5.1; p.24 onward	Exact cubic reflection and completion, with shared primes.
Lemma 7.1; pp.53–55	Exact local factors. Its printed second region needs extension.
Lemmas 8.1–8.2; pp.56–58;	Actual bins, witnesses, and a shared twist height; polynomial height
Prop.8.3; p.59 onward	ceiling.
Lemmas 10.3–10.6; pp.74–81	Printed boundary is $\sigma_0 \geq 7/8$. Reprove the required contour and remainder statements below that boundary.
Lemma 11.1; pp.82–83	Height chosen after all finite height orders, before the tail order.
Prop.11.3; pp.84–85	Transfer from the entire Hecke family to Dirichlet functions.
Lemma 14.3; pp.97–98;	Fixed-part reflected energy and Gram bound, with the stated masks and
Prop.15.2; pp.104–107	coefficient independence.
Prop.16.1; pp.110–112	Joint conductor allowance for all error slots, only in the permitted dynamic region.
Lemmas 17.1–17.2; pp.115–116	Marked inverse moment and its canonical input; strict widths, zero extensions, permitted row-independent coefficients.
Lemma 17.6; pp.151–152	Zero-slot sixth-power amplification; physical row valuations at most five.
Lemma 18.1; pp.153–154	Plain fourth moment. Use $\kappa = 3/4$ and baseline $\beta^* \leq 7/8$; retain inducing-family exclusions and slot mesh.
Lemma 19.1; p.181	Prime amplitudes and pointwise subdivisions under the shared height budget.

The following are new applications, not literal imports of printed numerical conclusions. Maintain one named proof obligation for each:

1. Define (12.5) with the new lengths and retain the quotient-free tuple correction (16.3), masks, and normalization.
2. Reprove Lemma 15.1 and Proposition 15.3 using the new rescaled row exponent and Appendix B’s low inequality.
3. Reprove Proposition 19.2 with fixed $\kappa = 3/4$. Its printed hypothesis $\beta^* > 7/8$ fails here. Derive the actual row count without its old $\Delta/4$ capacity loss, including small and zero capacities.
4. Extend the principal Euler region and slot estimate. Reprove the needed versions of Lemmas 10.3–10.6 below $7/8$: retain the nonprincipal region, extend the principal region, and use $\mathcal{D}_1(1/4)$ for small rows. Establish uniform bounds and a nonzero prime normalizer.
5. Recompute Lemma 20.1’s general exponent and all regimes in Appendix D. Do not import its old $7/8$ -specific expression.
6. Reprove the assembly of Section 20.6/Proposition 20.3 with the new margins and every quantifier in Appendix F.

An arithmetic certificate alone discharges none of these six analytic applications.

F Quantifier contract, height choice, and final contradiction

The source defines β^* as the supremum of $\{1/2\}$ together with the real parts in $[1/2, 1]$ of zeros of primitive finite-order Hecke L -functions over F . Formalize this actual set and its boundedness. The added $1/2$ ensures nonemptiness. A supremum need not be attained.

Assume $\sigma_0 < \beta^* \leq 7/8$ and put $\Delta = \beta^* - \sigma_0 > 0$. **Every positive gap must be handled.** Neither $\Delta \geq \gamma$ nor $\Delta \geq 21/500000$ is allowed. In this contradiction argument:

1. Before choosing a target character, fix the geometry, capacity losses, slot mesh and profiles, detector and amplitude widths, principal error losses, and absolute outer line. Choose common $0 < \omega < \Delta$ and $m > 0$. These may depend on Δ and the global family supremum.
2. For each target η , choose its excluded finite set, H_η , J_η , f_η , and finite orders $A_\eta \geq 0$, $B_\eta \in \mathbb{R}$, $\tau_{0,\eta} > 0$. The probe and signal are independent of later height and tail choices. Require H_η holomorphic and $|H_\eta(s) - 1| \leq 1/2$ on $\Re s > \sigma_0$.
3. Prove the low bound for all sufficiently large *real* Z , with character-dependent constants: $|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega}$.
4. For every integer $N \geq 0$ and every fixed $0 < \tau \leq \min(d_{\min}/100, \tau_{0,\eta})$, prove

$$|J_\eta(Z) - f_\eta(Z)| \ll_{\eta,N,\tau} Z^{C(\beta^*)-m}(1 + Z^\tau)^{A_\eta} + Z^{B_\eta - N\tau}.$$

Here $A_\eta, B_\eta, \tau_{0,\eta}$ were chosen *before* N . They cannot grow with the tail order. The implied constant and lower threshold for Z may depend on η, N, τ .

5. Choose $0 < \tau \leq \min(d_{\min}/100, \tau_{0,\eta}, m/[4(A_\eta + 1)])$, then N large enough that $B_\eta - N\tau < C(\beta^*) - m/2$. For $Z \geq 1$, $(1 + Z^\tau)^{A_\eta} \leq 2^{A_\eta} Z^{\tau A_\eta}$, so the high saving is the common $m/2 > 0$.

Use `Real.rpow` on positive real Z , or an explicit eventual bound $\exists C > 0 \exists Z_0 \geq 2 \forall Z \geq Z_0$. Integer-only scale bounds are insufficient for the continuous Mellin integral without a separate interpolation argument. The exact signal, with $C(s) = s - 11/16$, is

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{\Re s=2} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)} ds, \quad L_F^S = L_F \prod_{\mathfrak{p} \in S} (1 - \eta(\mathfrak{p}) N \mathfrak{p}^{-s}).$$

Proposition 2.1 uses $\varepsilon_* = \min(\Delta - \omega, m/2) > 0$ to continue all reciprocals to $\Re s > \beta^* - \varepsilon_*$. This boundary exceeds σ_0 . The least-upper-bound property supplies some target and zero with real part above $\beta^* - \varepsilon_*$; the added $1/2$ cannot be that witness. This contradicts holomorphic reciprocal continuation. Do not assume a rightmost zero or move a contour through an unexcluded pole.

Only then apply Proposition 11.3 to obtain the zeta conclusion. Setting the final target to zeta does not remove the moving internal Hecke characters.

G Mathlib zeta interface and honest theorem signatures

The `RiemannZeta` and `Nonvanishing` modules in mathlib’s L-series library expose the function `riemannZeta`. The latter contains `riemannZeta_ne_zero_of_one_le_re`. The current totalized value at $s = 1$ happens to be nonzero; it does not make zeta holomorphic there. Keep $s \neq 1$ in the main mathematical theorem. Do not substitute the zeros of the additive pole-removed function `completedRiemannZeta0` for zeta zeros.

```
-- Specification only; not compiler-checked in this handoff.
import Mathlib.NumberTheory.LSeries.Nonvanishing
import Mathlib.Tactic
namespace QuasiRiemann

def ZeroFreeRight (bnd : Real) : Prop :=
  forall s : Complex,
    bnd < s.re -> Not (s = 1) -> Not (riemannZeta s = 0)

def ZeroUpperBound (bnd : Real) : Prop :=
  forall s : Complex,
    Not (s = 1) -> riemannZeta s = 0 -> s.re <= bnd

theorem zeroFree_of_upperBound {bnd : Real}
  (hBound : ZeroUpperBound bnd) : ZeroFreeRight bnd := by
  intro s hs hPole hZero
  exact (not_le_of_gt hs) (hBound s hPole hZero)

end QuasiRiemann
```

This is merely a logical adapter. Supplying `hBound : ZeroUpperBound sigma0` would assume the improvement and would not formalize the argument. The actual conditional theorem should take the explicit source statements and prove the modified estimates, supremum bound, and transfer. Its intended shape is

```
-- Intended signature, not an admitted declaration:
-- theorem conditional_zero_free_improved
--   (inputs : SourceInputs) : ZeroFreeRight sigma0
```

Define every field of `SourceInputs` using actual functions and quantified estimates from Appendix E. It must not contain the desired improved bound, `False`, or an opaque “certificate implies zero-free” field. Parameter-modified steps must be proved as separate theorems, or explicitly listed as additional unproved hypotheses in a weaker conditional deliverable.

Useful small unconditional lemmas. Prove the late-height exponent choice in Appendix F independently of the automorphic objects. Likewise prove the order-theoretic lemma: a nonempty bounded set with supremum b has an element above every $c < b$. Then instantiate it with the actual zero set and exclude the adjoined $1/2$. These are legitimate components, not substitutes for the analytic estimates.

API provenance. Documentation checked on 8 October 2026: [RiemannZeta module](#) and [Nonvanishing module](#). Use the checked project’s pinned version to resolve precise names. No toolchain or mathlib revision has been tested for the listings in this PDF.

H Acceptance criteria and a ready-to-use Codex task

First acceptance target: arithmetic. Produce a clean Lean project with exact definitions and proofs of the geometry, low inequality, row balancing, uniform high margin, and auxiliary regime inequalities. Export a theorem named `arithmetic_parameter_certificate` bundling these facts. It must quantify over the full real rectangles and carry no analytic assumptions. Include the exact improvement $21/500000$.

Second target: explicit conditional analysis. Implement the actual source family, probe, Euler correction, and signal; formalize the modified applications in Appendix E. Export `conditional_zero_free_improved` with each unproved source input visible. If only a subset of the analytic assembly is achieved, report exactly that subset. Do not call an abstract zero-set wrapper a proof about zeta.

Build and trust report.

1. Pin `lean-toolchain` and the `mathlib` revision. Include the lockfile, build command, successful output, and source-PDF fingerprint.
2. Build every accepted module without `sorry`, `admit`, new project axioms, or unproved opaque propositions. Unfinished targets belong in a separate obligation list, not a file advertised as a proof.
3. Run `#check`, `#print`, and `#print axioms` on each exported theorem. Print all analytic input fields too. `#print axioms` alone does not reveal hypotheses passed as arguments.
4. Ordinary foundational axioms such as `propext`, `Classical.choice`, and `Quot.sound` are distinct from unproved analytic assertions. Report any `sorryAx` or new project axiom as a failure of an unconditional claim.
5. Check that the final type concerns the actual `riemannZeta`, all complex s with $\sigma_0 < \Re s$, and the stated pole exclusion. There must be no hidden finite-height restriction, rightmost-zero assumption, or lower bound on the hypothetical gap Δ .

Copy this task when handing the PDF to Codex:

Build a Lean 4/mathlib formalization of the exact 0.874958 candidate in this PDF. Start with Appendices B–D and complete the unconditional arithmetic certificate. Use rational constants and prove the polynomial identities and uniform bounds over real intervals; do not replace them by numerical tests. Then inspect the source manuscript identified in Appendix A and implement the analytic dependency plan in Appendices E–G. Keep all unproved analytic inputs explicit in the conditional theorem. Reprove the parameter-modified low bound, row count, Euler extension, and high assembly; do not import their old 7/8-specific versions verbatim. Follow the common-height quantifier order. Do not use an assumed improved zero bound, `sorry`, or a new axiom to claim an unconditional result. Deliver the compiled arithmetic modules, any completed conditional analysis, build logs, printed theorem types and axioms, and a precise list of remaining analytic obligations.

Reporting rule. A completed first target is an arithmetic certificate. A completed second target is a conditional theorem. Only after every analytic input has been discharged for the actual functions is the improved zeta zero-free theorem established in Lean.